# PAPER_684: UQFF Primordial Black Hole Evaporation: Suppressed Rate and Extended Lifetime

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**Framework:** Unified Quantum Field Framework (UQFF) v5.30
**Session:** 173
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**Class:** `UQFFPrimordialBHEvaporation` | CP4 #268
**Source:** `grok_share_fc21e30c24b4.txt`
**Domain:** Primordial Black Holes / Cosmology

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## Abstract

Primordial black holes evaporate via Hawking radiation with standard lifetime $\tau_{\text{std}}(M) = 5120\pi G^2 M^3/(\hbar c^4)$. In UQFF the evaporation rate is suppressed by the condensate factor: $\dot{M}_{\text{UQFF}} = \dot{M}_{\text{std}}(1 - f_{\text{TRZ}})(\rho_{\text{SCm}}/\rho_{\text{UA}})e^{-U_m/k_B T_H}$, extending the PBH lifetime by a factor $\tau_{\text{UQFF}}/\tau_{\text{std}} \approx \rho_{\text{UA}}/((1 - f_{\text{TRZ}})\rho_{\text{SCm}}) \approx 11$.

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## Primary UQFF Equation

$$\dot{M}_{\text{UQFF}} = -\frac{\hbar c^4}{15360\pi G^2 M^2}(1 - f_{\text{TRZ}})\frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}}e^{-U_m/k_B T_H}$$

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## Parameters

| Parameter | Value | Description |
|---|---|---|
| $M_0$ | $10^{12}$ kg | Initial PBH mass |
| $\tau_{\text{UQFF}}/\tau_{\text{std}}$ | $\sim 11$ | Lifetime extension |
| $t_{\text{form}}$ | $10^{-23}$ s | Formation epoch |

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## UQFF Constants

| Constant | Value | Description |
|---|---|---|
| $\rho_{\text{UA}}$ | $7.09 \times 10^{-36}$ kg/m$^3$ | Universal Aether density |
| $\rho_{\text{SCm}}$ | $7.09 \times 10^{-37}$ kg/m$^3$ | Schwarzschild condensate |
| $f_{\text{TRZ}}$ | 0.1 | Time-reversal zone factor |
| $\kappa$ | $5 \times 10^{-4}$ day$^{-1}$ | UQFF calibration constant |
| $[\text{SSq}]$ | 0.57 | Superstring quenching factor |
| $\mu_J$ | $3.38 \times 10^{23}$ J·m | Magnetic string coupling |
| $\gamma$ | $5 \times 10^{-5}$ / 86400 s$^{-1}$ | Aether oscillation decay |

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### C++ Implementation

```cpp
#include "UQFFPrimordialBHEvaporation.h"

UQFFPrimordialBHEvaporation obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

### Python CP4 Calculator

```python
from CondensedPhysics2 import UQFFPrimordialBHEvaporationCalculator

calc = UQFFPrimordialBHEvaporationCalculator()
result = calc.compute()  # returns all UQFF quantities
print(result)
```

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### References


Hawking 1974; Carr et al. 2016 (PBH review); Green et al. 2021; UQFF Session 173


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